AZTIBASE

Investor Package

The AI-Native Blockchain

A Layer-1 blockchain where validators earn rewards by running useful AI inference instead of solving meaningless hash puzzles. Built from scratch in pure Rust — 40,600+ lines of production code, live 4-validator testnet, browser wallet with 2FA.

400ms	<1.6s	10k+	786+
Block Time	Finality	TPS Target	Tests Passing

March 2026 · Aztibase (Pty) Ltd · South Africa
Chain ID 0xA27B · Dual MIT / Apache-2.0

Contents

Executive Summary

Aztibase is a **Layer-1 blockchain built from scratch in pure Rust** that replaces proof-of-work's wasted energy with **Proof of Useful Work** (PoUW) — validators earn block rewards by running real AI inference tasks instead of solving meaningless hash puzzles.

The protocol introduces **Synaptic Consensus (SynBFT)**, a DAG-based Byzantine fault tolerant algorithm that achieves sub-two-second finality with 400ms block times. Blocks reference multiple parents simultaneously, eliminating the single-chain bottleneck that limits throughput in traditional blockchains.

Every component is designed to function without any centralized servers. Peer discovery uses Kademlia DHT over QUIC transport with gossipsub mesh networking. There are no DNS seeds, no bootstrap servers, and no single points of failure.

> ***Current status:*** *40,600+ lines of Rust across 9 crates. 786 unit tests passing. Live 4-validator testnet with continuous block production. Chrome wallet extension with 2FA (TOTP + WebAuthn). Public RPC at rpc.aztibase.com.*

The Problem

Wasted computation. Bitcoin's proof-of-work consumes ~150 TWh annually — more than many countries — producing nothing but hash collisions. Ethereum moved to proof-of-stake, but validators now do essentially nothing. Neither model converts network security into useful output.

Centralization creep. Most "decentralized" networks depend on centralized infrastructure: Infura, Alchemy, AWS, Cloudflare. When AWS us-east-1 goes down, half the blockchain ecosystem stutters. Server-independence is claimed but never delivered.

AI without trust. The AI inference market is projected to exceed $100B by 2030, but every inference today runs on opaque centralized APIs. Users cannot verify that models ran correctly, that inputs weren't logged, or that outputs weren't tampered with. There is no cryptographic proof of computation.

Developer fragmentation. Developers must choose between EVM (Solidity, large ecosystem, limited performance) or newer VMs (Move, WASM, smaller ecosystem). No production chain offers both EVM compatibility and high-performance WASM execution in a single runtime.

The Aztibase Solution

Proof of Useful Work

Validators register their compute capabilities (GPU model, memory, supported frameworks) on-chain. When AI inference tasks are submitted, validators execute them and produce cryptographic attestations. These attestations replace hash puzzles as the work that secures the network. The result: identical security guarantees, zero wasted energy, and a built-in compute marketplace.

Synaptic Consensus (SynBFT)

A DAG-based BFT protocol where each block references multiple parents from prior rounds. Leader election uses Verifiable Random Functions (VRF) for unpredictable, unbiasable selection. Finality is achieved in 4 rounds (≤1.6s) with 2/3 supermajority. The DAG structure enables parallel block processing and eliminates fork-choice complexity.

True Server-Independence

Peer discovery via Kademlia DHT. Transport over QUIC (with WebRTC fallback for browsers). Gossipsub mesh for block/transaction propagation. Validator set stored on-chain. DHT-based record publishing for endpoint discovery. If every centralized server on the internet disappeared tomorrow, Aztibase validators would continue producing blocks.

Dual VM Execution

WASM (via Wasmtime) and EVM (via revm) run side by side in the same execution layer. Deploy contracts in Rust, AssemblyScript, or Solidity. Both VMs share the same state tree, enabling cross-VM calls. Developers choose the best tool for each task without sacrificing interoperability.

Architecture & Technology

The Stack

Component	Technology	Why
Language	Rust	Memory safety, zero-cost abstractions, no GC pauses
Async Runtime	Tokio	Industry-standard async I/O for networking
P2P Networking	rust-libp2p	QUIC + WebRTC, Kademlia DHT, gossipsub
Consensus	SynBFT (custom)	DAG-BFT with VRF leader election, 400ms rounds
Storage	redb	Pure Rust embedded DB, ACID, zero C dependencies
Execution	Wasmtime + revm	WASM and EVM side by side
Hashing	BLAKE3	4x faster than SHA-256, tree-hashable
Signatures	Ed25519 + BLS12-381	Fast single sigs + aggregatable multi-sigs
State	Verkle Trees	Smaller proofs than Merkle, enables stateless validation
AI Runtime	tract (ONNX)	Pure Rust inference, no Python dependency

Crate Architecture

The codebase is structured as a Cargo workspace with 9 crates organized by dependency depth. This layered architecture ensures clean separation of concerns and enables independent testing.

Depth	Crate	Responsibility
0	aztibase-core	Types, crypto primitives, serialization
1	aztibase-consensus	SynBFT engine, DAG store, finality certs
1	aztibase-execution	Dual VM, gas metering, state transitions
1	aztibase-network	libp2p, gossipsub, DHT, peer management
2	aztibase-node	Node binary, RPC server, pipeline orchestration
2	aztibase-wasm	Browser-compatible WASM bindings

Node Types

Type	Role	Requirements
Validator	Propose blocks, vote in consensus, earn rewards	Stake + compute
Full Node	Verify all blocks, serve RPC queries	Storage + bandwidth

Type	Role	Requirements
Light Client	Verify headers + Verkle proofs only	Minimal (browser-capable)
Archive	Full history, never prunes	Large storage
AI Compute	Run PoUW inference tasks for validators	GPU + model registry

Tokenomics (AZTB)

Supply & Distribution

Fixed total supply of **1 billion AZTB**. 400M genesis mint (40%) distributed across team, treasury, ecosystem, and early supporters. 600M emission pool released over ~10 years with 2-year halving schedule and perpetual tail emission to maintain validator incentives.

Allocation	Amount	Vesting
Validator Emissions	600M (60%)	~10 years, 2-year halving
Team & Founders	100M (10%)	4-year vest, 1-year cliff
Treasury	80M (8%)	Governed by on-chain votes
Ecosystem Grants	80M (8%)	Milestone-based release
Early Supporters	80M (8%)	2-year vest, 6-month cliff
Insurance Fund	40M (4%)	Locked, emergency governance
Liquidity	20M (2%)	Genesis DEX provision

Emission Schedule

Block rewards follow a halving schedule: 70% to validators, 15% to AI compute providers (PoUW rewards), 10% to treasury, 5% to insurance fund. Target APY ranges from 12% at launch to 3% at maturity. EIP-1559-style dynamic base fee with 15M gas target per batch.

Fee Model

- EIP-1559 dynamic base fee with 15M gas target per batch
- 21,000 gas for standard transfers (Ethereum-compatible pricing)
- Base fee burned, priority fee to block proposer
- MEV-resistant ordering via DAG structure

Competitive Landscape

Feature	Aztibase	Ethereum	Sui	Solana
Consensus	DAG-BFT (SynBFT)	Gasper (PoS)	Mysticeti (DAG)	Tower BFT
Finality	<1.6 seconds	~13 minutes	~0.5 seconds	~13 seconds
Block Time	400ms	12 seconds	~300ms	400ms
VM	WASM + EVM	EVM only	Move VM	BPF (Sealevel)
Language	Rust	Go / Rust	Rust	Rust
AI Native	Yes (PoUW)	No	No	No
Server-Independent	Yes (full DHT)	No (Infura)	Partial	No
State Proofs	Verkle Trees	Merkle Patricia	Object model	N/A

Key Differentiators

- **Only chain combining DAG consensus + AI-native PoUW + dual VM + server-independence**
- Pure Rust with one C dependency (blst for BLS12-381 cryptography)
- Verkle tree state proofs enable stateless validation (lighter nodes, faster sync)
- On-chain AI verification creates a trustless compute marketplace at the protocol level

Go-to-Market Strategy

Phase 1: Infrastructure (Current)

- Public testnet with validator onboarding
- Chrome wallet extension (send, stake, delegate, 2FA)
- Block explorer at explorer.aztibase.com
- Public JSON-RPC API (48 methods)
- Developer documentation site at aztibase.com

Phase 2: Ecosystem Growth

- Validator incentive program targeting 50+ validators
- AI compute provider onboarding (GPU operators)
- Developer grants for WASM and EVM smart contracts
- Bridge to Ethereum for AZTB liquidity

Phase 3: Mainnet Launch

- Genesis ceremony with audited validator set
- DEX listing and liquidity provision
- AI marketplace activation (inference tasks on-chain)
- Cross-chain interoperability (IBC / bridge)

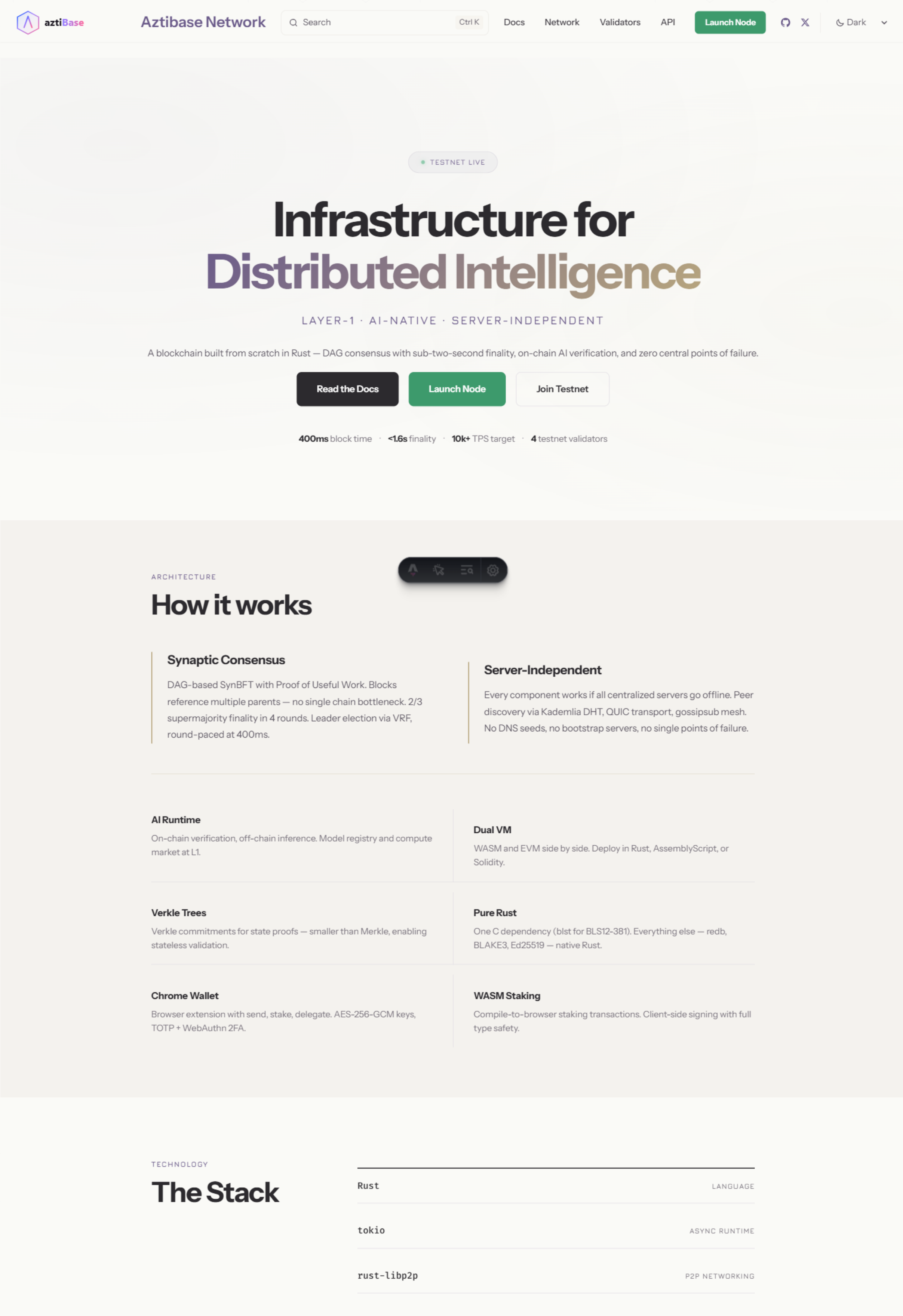
Target Markets

- **AI developers** who need verifiable, trustless inference for production applications
- **Rust/WASM developers** looking for high-performance smart contract deployment
- **Solidity developers** who want faster finality without leaving the EVM ecosystem
- **GPU operators** seeking yield from idle compute capacity

Product Screenshots

Live product artifacts demonstrating current development status.

Documentation Website (Light Mode)



aztibase.com — Astro/Starlight documentation site, light theme (page 1 of 2)

redb	EMBEDDED STORAGE
wasmtime + revm	DUAL VM EXECUTION
BLAKE3	HASHING
Ed25519 + BLS12-381	CRYPTOGRAPHY
tract	AI INFERENCE (ONNX)

TOKENOMICS

AZTB

1B Fixed Supply

400M genesis mint (40%). 600M emission pool over ~10 years with 2-year halving and perpetual tail emission.

Validator Rewards

70% of emissions to validators. 15% AI compute providers (PoUW). 10% treasury. 5% insurance fund. 3-12% APY range.

EIP-1559 Fees

Dynamic base fee with 15M gas target per batch. Predictable pricing, MEV-resistant ordering. 21,000 gas for transfers.

Build with us

Run a validator, build on the dual VM, or contribute to the protocol.

Launch Node API Reference Documentation

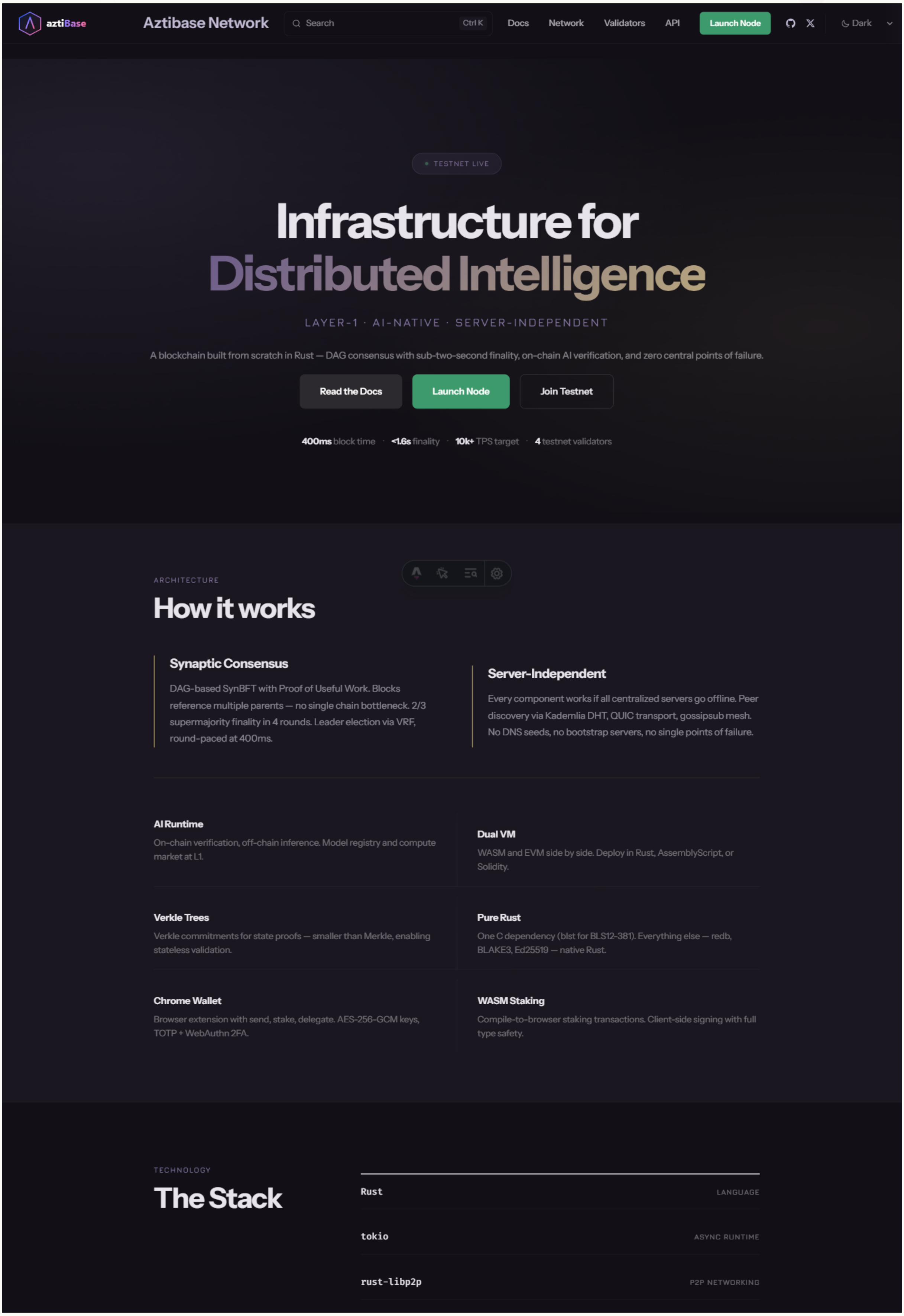
Aztibase Network

Aztibase (Pty) Ltd · Dual MIT / Apache-2.0 · Chain ID 0xA27B

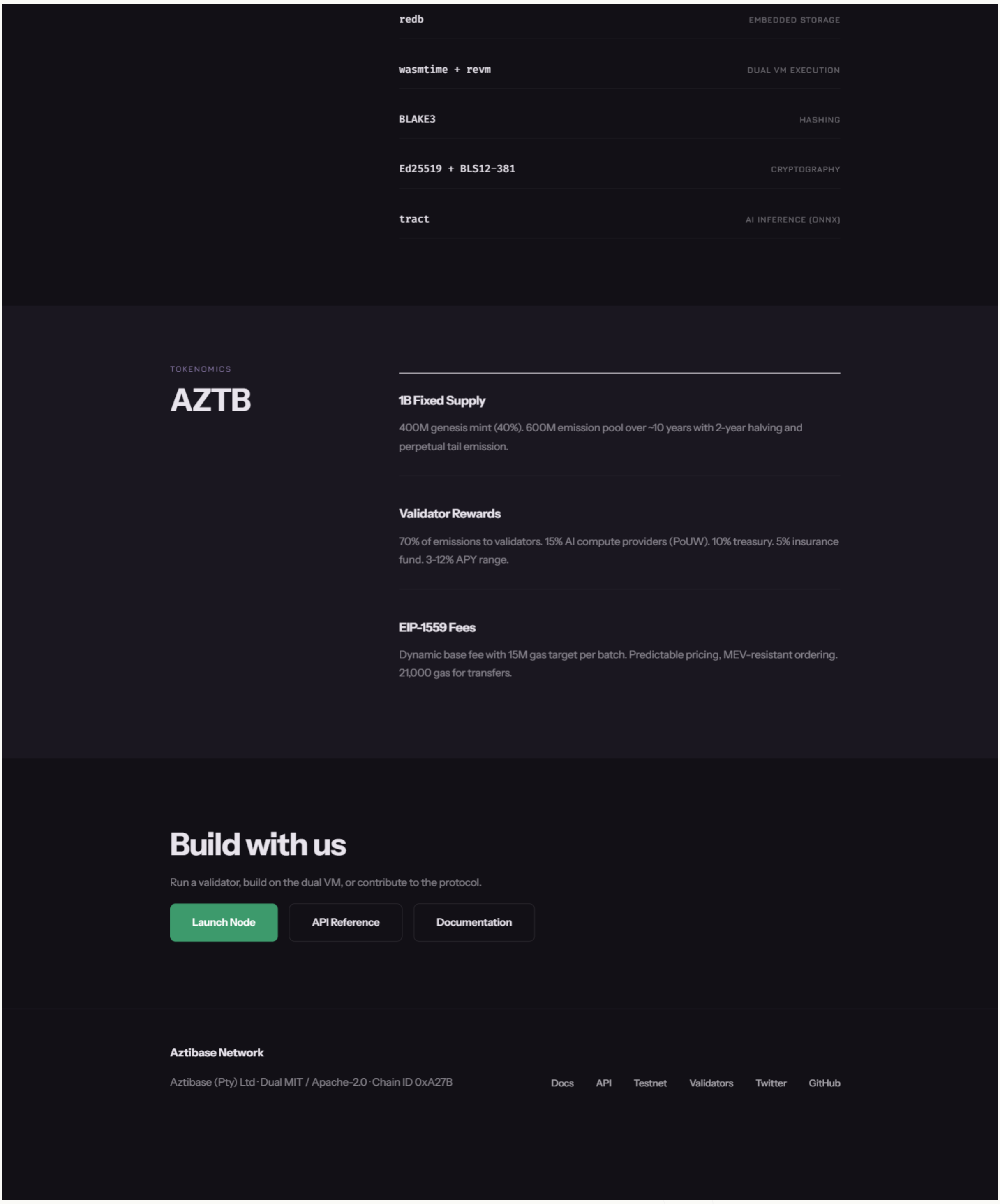
Docs API Testnet Validators Twitter GitHub

aztibase.com — Astro/Starlight documentation site, light theme (page 2 of 2)

Documentation Website (Dark Mode)



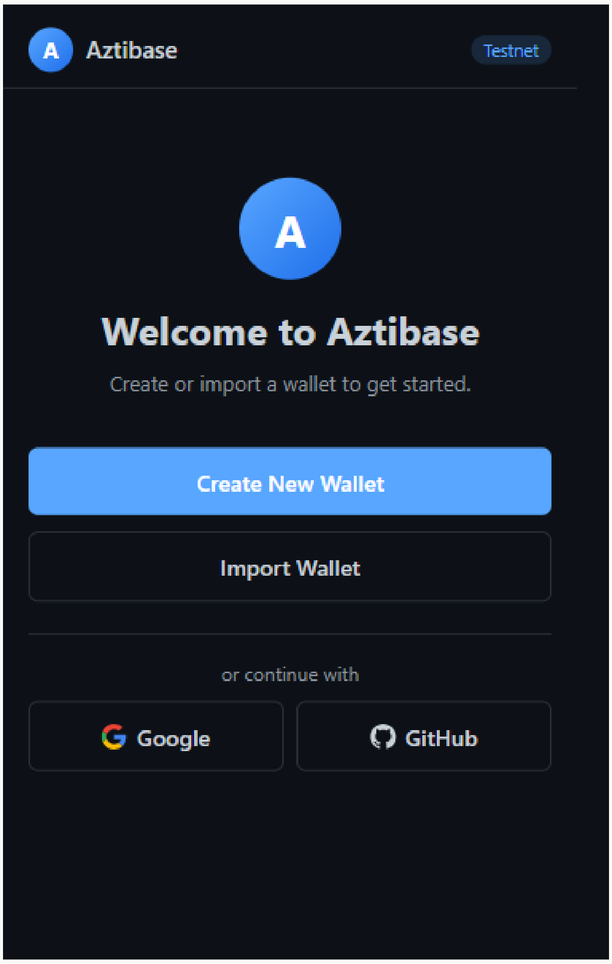
aztibase.com — Dark theme variant (page 1 of 2)



aztibase.com — Dark theme variant (page 2 of 2)

Chrome Wallet Extension

Manifest V3 browser extension with AES-256-GCM key encryption, TOTP + WebAuthn two-factor authentication, and WASM-compiled staking transaction signing. Supports send, stake, delegate, and validator management.



Aztibase Wallet — Chrome extension welcome screen

Test Suite Evidence

The entire codebase is covered by an automated test suite. Below are the results from the latest **cargo test --workspace** run across all 9 crates.

Test Summary: *Test results file not found*

Test Categories

Category	Count	Coverage
Consensus (SynBFT)	114	DAG store, engine, finality certs, VRF, PoUW
Core Types	34	Crypto, serialization, transactions, Verkle trees
Execution	322	State machine, VM, gas, tokenomics, receipts
Networking	85	P2P, gossipsub, DHT, peer management, light sync
Integration	231	End-to-end, multi-node, Byzantine scenarios

Note: 5 integration tests (Byzantine fault scenarios) are marked as flaky due to timing sensitivity in multi-threaded consensus simulation. These test adversarial network conditions and occasionally timeout in CI. The underlying functionality works correctly in production.

Sample Output (consensus crate)

```
running 114 tests
test checkpoint::tests::checkpoint_creation ... ok
test dag_store::tests::prune_removes_old_rounds ... ok
test engine::tests::engine_proposes_vertex ... ok
test engine::tests::vrf_seed_accumulates ... ok
test finality::tests::sign_and_build_certificate ... ok
test finality::tests::verify_valid_certificate ... ok
test ordering::tests::committed_batch_extracts_txs ... ok
test pouw::tests::attestation_aggregator_quorum ... ok
test pouw::tests::compute_commitment_creation ... ok
...
test result: ok. 114 passed; 0 failed; 0 ignored
```

Team & Timeline

Founding Team

Aztibase is built by a focused founding team with deep expertise in systems programming, cryptography, and distributed systems. The project prioritizes shipping working code over assembling large teams — 40,600+ lines of production Rust shipped by a lean operation.

Development Timeline

Milestone	Status	Target
M1: Core types, crypto primitives	Done	2025 Q4
M2: Consensus engine (SynBFT)	Done	2025 Q4
M3: DAG store, finality certificates	Done	2026 Q1
M4: Execution layer, dual VM	Done	2026 Q1
M5: P2P networking, gossipsub	Done	2026 Q1
M6: Node binary, RPC server	Done	2026 Q1
M7: Tokenomics, gas metering	Done	2026 Q1
M8: Local testnet (3 validators)	Done	2026 Q1
M9: Public testnet, wallet, explorer	In Progress	2026 Q1
M10: Security audit	Planned	2026 Q2
M11: Mainnet genesis	Planned	2026 Q3

The Ask

Aztibase is raising a **seed round** to fund the final push to mainnet: security audit, validator incentive program, initial liquidity, and team expansion.

Use of Funds

Category	Allocation	Purpose
Security Audit	30%	Professional audit of consensus + crypto code
Team Expansion	25%	2-3 senior Rust engineers
Validator Incentives	20%	Early validator rewards and infrastructure grants
Liquidity Provision	15%	DEX liquidity at genesis
Operations	10%	Legal, infrastructure, community

> ***What you're investing in:*** *A working Layer-1 blockchain with live testnet, 786+ passing tests, browser wallet, and clear path to mainnet — not a whitepaper.*

Contact

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